

INFLAMMATORY, DEGENERATIVE & CONNECTIVE-TISSUE DISORDERS

20-Mark University Examination Answers — Orthopaedics / BPT

The following answers are arranged in a 20-mark university format with emphasis on definition, causes/etiology, clinical features, complications, deformities, radiological features, conservative management, surgical management and physiotherapy where relevant.

1. OSTEOARTHRITIS

DEFINITION

Osteoarthritis (OA) is a chronic, progressive degenerative disorder of synovial joints characterized by loss of articular cartilage, subchondral bone changes, osteophyte formation and varying degrees of synovial inflammation, resulting in pain, stiffness and functional limitation.

CAUSES / RISK FACTORS

Primary OA

Occurs without a single identifiable cause.

Risk factors:

- Increasing age
- Female sex
- Obesity
- Genetic predisposition
- Previous joint injury
- Repetitive joint loading

Secondary OA

Occurs due to an identifiable condition:

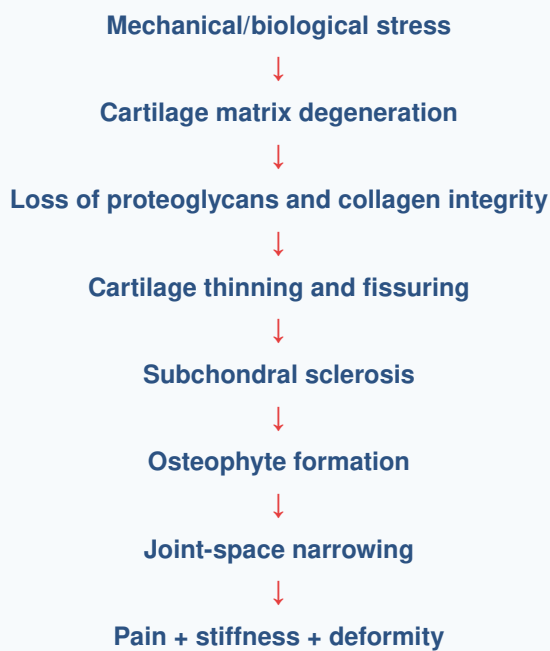
- Previous fracture
- Joint infection
- Rheumatoid arthritis
- Congenital joint abnormality
- Ligament injury
- Meniscal injury
- Avascular necrosis
- Metabolic disorders

COMMON SITES

- Knee
- Hip
- Spine
- Hands
- First MTP joint

- Shoulder and other joints less commonly

PATHOPHYSIOLOGY



CLINICAL FEATURES

Symptoms

- Joint pain, usually worse with activity
- Pain relieved by rest initially
- Morning stiffness usually less than 30 minutes
- Stiffness after prolonged inactivity
- Difficulty walking
- Difficulty climbing stairs
- Reduced functional activity

Signs

- Joint tenderness
- Crepitus
- Reduced ROM
- Bony enlargement
- Joint-line tenderness
- Muscle weakness
- Deformity in advanced disease

Hand OA

- Heberden's nodes — DIP joints
- Bouchard's nodes — PIP joints

DEFORMITIES

Knee

- Varus deformity
- Valgus deformity

- Flexion deformity

Hip

- Flexion deformity
- External rotation
- Adduction deformity

Hands

- Heberden's and Bouchard's nodes

RADIOLOGICAL FEATURES

X-ray shows:

- Non-uniform joint-space narrowing
- Osteophytes
- Subchondral sclerosis
- Subchondral cysts
- Bony remodeling
- Deformity in advanced disease

Classic mnemonic: LOSS

- Loss of joint space
- Osteophytes
- Subchondral sclerosis
- Subchondral cysts

COMPLICATIONS

- Severe pain
- Joint deformity
- Contractures
- Muscle wasting
- Reduced mobility
- Disability
- Secondary instability
- Reduced quality of life

CONSERVATIVE MANAGEMENT

Medical

- Patient education
- Weight reduction
- Activity modification
- Paracetamol/analgesics where appropriate
- NSAIDs when appropriate
- Topical NSAIDs
- Intra-articular corticosteroid injection in selected cases
- Management of comorbidities

Physiotherapy

- ROM exercises
- Quadriceps strengthening

- Hip strengthening
- Aerobic exercise
- Aquatic exercise
- Gait training
- Balance exercises
- Assistive devices
- Bracing when indicated

SURGICAL MANAGEMENT

Indicated in severe symptomatic disease with significant functional limitation despite conservative treatment.

Procedures

- Arthroscopic procedures in selected circumstances
- Osteotomy in selected younger patients with appropriate deformity
- Total knee replacement
- Total hip replacement
- Other joint replacement procedures

2. RHEUMATOID ARTHRITIS

DEFINITION

Rheumatoid arthritis (RA) is a chronic systemic autoimmune inflammatory disease characterized primarily by persistent inflammatory synovitis of multiple joints, leading to cartilage destruction, bone erosion, deformity and extra-articular manifestations.

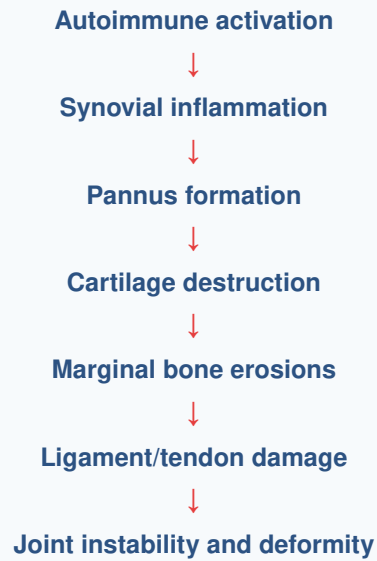
CAUSES / RISK FACTORS

Exact cause is unknown.

Important factors:

- Genetic predisposition
- Autoimmune mechanisms
- Female sex
- Smoking
- Environmental factors
- Hormonal factors
- Certain infections may act as triggers

PATHOPHYSIOLOGY



CLINICAL FEATURES

General

- Fatigue
- Malaise
- Fever
- Loss of appetite
- Weight loss

Joint

- Symmetrical polyarthritis
- Pain
- Swelling
- Warmth
- Prolonged morning stiffness, typically >30–60 minutes
- Reduced ROM

Commonly affects:

- MCP
- PIP
- Wrist
- MTP
- Knee
- Elbow
- Shoulder
- DIP joints are usually spared.

CHARACTERISTIC DEFORMITIES

Hand

- Ulnar deviation
- Swan-neck deformity

- Boutonnière deformity
- Z-thumb deformity
- Wrist deformity

Foot

- Hallux valgus
- MTP subluxation
- Claw toes

Cervical spine

- Atlantoaxial instability/subluxation

COMPLICATIONS

Joint

- Destruction
- Subluxation
- Ankylosis
- Deformity
- Disability

Extra-articular

- Rheumatoid nodules
- Vasculitis
- Interstitial lung disease
- Pleural disease
- Pericarditis
- Eye involvement
- Anemia
- Osteoporosis
- Increased cardiovascular risk

RADIOLOGICAL FEATURES

Early:

- Soft-tissue swelling
- Periarticular osteopenia

Later:

- Uniform joint-space narrowing
- Marginal erosions
- Subluxation
- Deformity
- Ankylosis in advanced disease

Cervical spine:

- Atlantoaxial instability
- Erosive changes

CONSERVATIVE MANAGEMENT

Medical

Early treatment with DMARDs is central.

May include:

- Methotrexate
- Hydroxychloroquine
- Sulfasalazine
- Leflunomide
- Biological/targeted DMARDs
- NSAIDs
- Corticosteroids for selected indications

Physiotherapy

- Pain relief
- ROM exercises
- Muscle strengthening
- Joint protection
- Splinting
- Energy conservation
- Hydrotherapy
- Functional training
- Assistive devices

SURGICAL MANAGEMENT

Indications:

- Severe joint destruction
- Persistent pain
- Functional disability
- Deformity

Procedures:

- Synovectomy
- Tendon repair
- Arthrodesis
- Joint replacement
- Cervical stabilization when required

3. ANKYLOSING SPONDYLITIS

DEFINITION

Ankylosing spondylitis (AS) is a chronic inflammatory rheumatic disease predominantly affecting the axial skeleton and sacroiliac joints, causing inflammatory back pain, progressive spinal stiffness and, in advanced disease, spinal fusion. It belongs to the axial spondyloarthritis spectrum.

CAUSES / RISK FACTORS

- Strong genetic association with HLA-B27

- Family history
- Immune-mediated mechanisms
- Environmental factors

CLINICAL FEATURES

Early

- Insidious low-back pain
- Buttock pain
- Sacroiliac pain
- Morning stiffness
- Improvement with exercise
- Worsening with rest
- Night pain

Progressive

- Reduced spinal mobility
- Chest expansion
- Hip involvement
- Fatigue

DEFORMITIES

Advanced disease may produce:

- Loss of lumbar lordosis
- Increased thoracic kyphosis
- Forward head posture
- Hip flexion deformity
- Fixed spinal posture

Classically described as a “bamboo spine” in advanced radiographic disease.

COMPLICATIONS

- Severe spinal stiffness
- Hip arthritis
- Osteoporosis
- Vertebral fractures
- Spinal cord injury after fracture
- Uveitis
- Cardiovascular involvement
- Restrictive chest wall mechanics
- Functional disability

RADIOLOGICAL FEATURES

X-ray

- Sacroiliitis
- Vertebral squaring
- Vertebral corner erosions
- Syndesmophytes
- Bridging of vertebrae

- Bamboo spine
- Hip joint changes

MRI

Can demonstrate:

- Active sacroiliitis
- Bone marrow edema
- Early inflammatory changes

CONSERVATIVE MANAGEMENT

Medical

- NSAIDs
- Biologic agents such as TNF inhibitors or IL-17 inhibitors when indicated
- Other targeted therapy in selected patients

Physiotherapy

Very important:

- Spinal mobility exercises
- Postural correction
- Chest expansion exercises
- Breathing exercises
- Strengthening
- Hydrotherapy
- Stretching
- Aerobic conditioning
- Patient education
- Maintain upright posture
- Avoid prolonged flexed posture
- Regular exercise

SURGICAL MANAGEMENT

Indications:

- Severe hip destruction
- Fixed severe deformity
- Neurological compression
- Unstable spinal fracture

Procedures:

- Total hip replacement
- Corrective spinal osteotomy in selected severe deformities
- Spinal stabilization/decompression when indicated

4. GOUTY ARTHRITIS

DEFINITION

Gout is an inflammatory arthritis caused by deposition of monosodium urate crystals in joints and tissues as a result of persistent hyperuricemia.

CAUSES / RISK FACTORS

Primary

Usually related to impaired uric acid excretion or increased production.

Secondary

- Chronic kidney disease
- Myeloproliferative disorders
- Certain medications
- Tumour lysis
- High cell turnover

Risk factors:

- Obesity
- Alcohol excess
- High-purine diet
- Metabolic syndrome
- Renal disease
- Family history

COMMON SITES

- First MTP joint — podagra
- Ankle
- Knee
- Midfoot
- Elbow
- Wrist
- Fingers

CLINICAL FEATURES

Acute attack

- Sudden severe pain
- Redness
- Swelling
- Warmth
- Extreme tenderness
- Often occurs at night

Chronic gout

- Recurrent attacks
- Tophi
- Joint destruction
- Chronic stiffness

Tophi

Deposits may occur in:

- Ear
- Fingers
- Toes

- Elbow
- Achilles tendon

COMPLICATIONS

- Chronic arthritis
- Tophi
- Joint destruction
- Kidney stones
- Chronic kidney disease
- Reduced function

DEFORMITIES

- Irregular joint enlargement
- Destructive deformity
- Tophi-related deformity

RADIOLOGICAL FEATURES

X-ray:

- Well-defined punched-out erosions
- Overhanging edges
- Preserved joint space until later disease
- Soft-tissue tophi
- Bone destruction in advanced disease

CONSERVATIVE MANAGEMENT

Acute attack

- NSAIDs
- Colchicine
- Corticosteroids
- Rest and ice
- Avoid joint trauma

Long-term

- Weight management
- Reduce alcohol
- Dietary modification
- Adequate hydration
- Urate-lowering therapy:
 - Allopurinol
 - Febuxostat in appropriate patients
- Treatment should be individualized based on renal function and comorbidities.

SURGICAL MANAGEMENT

Usually not required.

May be indicated for:

- Large symptomatic tophi
- Nerve compression

- Severe joint destruction
- Recurrent infection over tophi
- Severe functional impairment

Procedures:

- Tophus excision
- Joint reconstruction/replacement in selected cases

5. PSORIATIC ARTHRITIS

DEFINITION

Psoriatic arthritis (PsA) is a chronic inflammatory musculoskeletal disease associated with psoriasis, involving peripheral joints, the axial skeleton, entheses and digits.

CAUSES / RISK FACTORS

- Genetic predisposition
- Psoriasis
- Immune dysregulation
- Family history
- Environmental factors

CLINICAL FEATURES

Joint

- Asymmetrical arthritis
- Pain
- Swelling
- Stiffness
- Morning stiffness

Characteristic

- Dactylitis — “sausage digit”
- Enthesitis
- Commonly:
 - Achilles tendon
 - Plantar fascia

Skin

- Psoriatic plaques
- Nail pitting
- Nail dystrophy

Spine

- Inflammatory back pain in axial disease

DEFORMITIES

- Joint deformity
- Flexion deformities
- Pencil-in-cup deformity in severe disease
- Arthritis mutilans

- Ankylosis

COMPLICATIONS

- Permanent joint damage
- Severe functional disability
- Spinal stiffness
- Eye inflammation
- Cardiovascular/metabolic comorbidity

RADIOLOGICAL FEATURES

- Erosions
- New bone formation
- Asymmetric joint-space loss
- Periostitis
- Pencil-in-cup deformity
- Ankylosis
- Sacroiliitis
- Syndesmophytes

CONSERVATIVE MANAGEMENT

Medical

- NSAIDs
- DMARDs:
 - Methotrexate
 - Sulfasalazine in selected peripheral disease
- Biologic agents
- Targeted synthetic DMARDs
- Treatment of psoriasis

Physiotherapy

- ROM exercises
- Strengthening
- Hydrotherapy
- Postural exercises
- Joint protection
- Functional training

SURGICAL MANAGEMENT

For advanced joint destruction:

- Synovectomy in selected cases
- Arthrodesis
- Joint replacement
- Tendon procedures

6. HAEMOPHILIC ARTHRITIS

DEFINITION

Haemophilic arthritis is a chronic destructive arthropathy caused by recurrent bleeding into joints (hemarthrosis) in patients with haemophilia.

It commonly affects:

- Knee
- Ankle
- Elbow

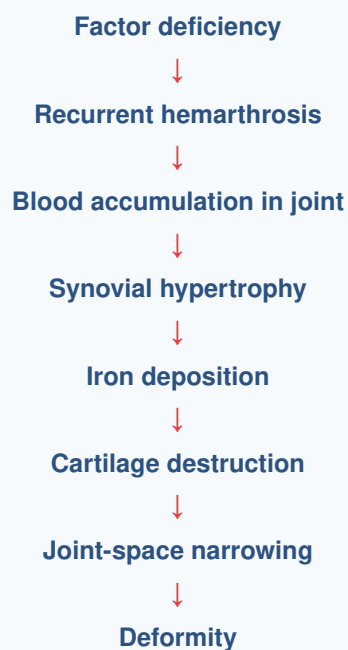
CAUSE

Usually occurs in:

- Hemophilia A — factor VIII deficiency
- Hemophilia B — factor IX deficiency

Repeated hemarthroses produce chronic synovial inflammation and cartilage destruction.

PATHOPHYSIOLOGY



CLINICAL FEATURES

Acute hemarthrosis

- Sudden joint pain
- Swelling
- Warmth
- Restricted movement
- Protective muscle spasm

Chronic

- Recurrent swelling
- Chronic pain
- Stiffness

- Muscle wasting
- Flexion contractures
- Reduced ROM

DEFORMITIES

- Knee flexion deformity
- Elbow flexion deformity
- Ankle equinus
- Joint enlargement
- Muscle wasting

COMPLICATIONS

- Chronic arthropathy
- Contractures
- Joint deformity
- Muscle wasting
- Secondary osteoarthritis
- Severe disability
- Recurrent bleeding

RADIOLOGICAL FEATURES

Early:

- Soft-tissue swelling
- Joint effusion

Advanced:

- Epiphyseal enlargement
- Subchondral cysts
- Joint-space narrowing
- Osteophytes
- Erosions
- Irregular articular surfaces

MRI can identify:

- Synovial hypertrophy
- Hemosiderin deposition
- Cartilage damage

CONSERVATIVE MANAGEMENT

Acute bleed

Factor replacement is the main treatment.

Additional:

- Rest
- Ice
- Compression where appropriate
- Elevation
- Analgesia

- Protected movement initially

Long-term

- Factor prophylaxis where indicated

Physiotherapy

- Muscle strengthening
- ROM exercises
- Joint protection
- Weight management

SURGICAL MANAGEMENT

For advanced arthropathy:

- Synovectomy
- Joint replacement
- Arthroscopic procedures in selected cases
- Corrective procedures for severe deformity

Adequate factor replacement is essential around surgery.

7. STILL'S DISEASE / JUVENILE IDIOPATHIC ARTHRITIS

DEFINITION

Systemic juvenile idiopathic arthritis (Still's disease) is a subtype of juvenile idiopathic arthritis characterized by arthritis with systemic inflammation, fever and often an evanescent salmon-pink rash.

CAUSES

Exact cause is unknown.

Possible mechanisms:

- Autoinflammatory immune dysfunction
- Genetic predisposition
- Environmental triggers

CLINICAL FEATURES

Systemic

- High spiking fever
- Evanescent salmon-pink rash
- Fatigue
- Malaise
- Lymphadenopathy
- Hepatosplenomegaly
- Serositis

Joint

- Pain
- Swelling
- Stiffness
- Reduced ROM

- Polyarthritis

DEFORMITIES

Untreated chronic disease may produce:

- Flexion deformities
- Growth disturbance
- Limb-length discrepancy
- Joint destruction
- Contractures

COMPLICATIONS

- Growth retardation
- Joint destruction
- Contractures
- Osteoporosis
- Macrophage activation syndrome
- Chronic disability
- Amyloidosis in longstanding inflammatory disease

RADIOLOGICAL FEATURES

Early:

- Soft-tissue swelling
- Osteopenia

Later:

- Joint-space narrowing
- Growth abnormalities
- Erosions
- Deformity

CONSERVATIVE / MEDICAL MANAGEMENT

- NSAIDs
- Corticosteroids
- Methotrexate for persistent arthritis
- Biologic agents, especially IL-1 or IL-6 targeted therapy where indicated
- Physiotherapy
- Occupational therapy
- Nutritional support

SURGICAL MANAGEMENT

Rarely required in active systemic disease.

In severe chronic joint destruction:

- Synovectomy in selected cases
- Joint replacement
- Corrective surgery

8. CHARCOT JOINT

DEFINITION

Charcot joint (neuropathic arthropathy) is a progressive destructive arthropathy resulting from loss of protective sensation and proprioception, leading to repetitive microtrauma, joint instability, deformity and destruction.

CAUSES

Most common:

- Diabetes mellitus with peripheral neuropathy

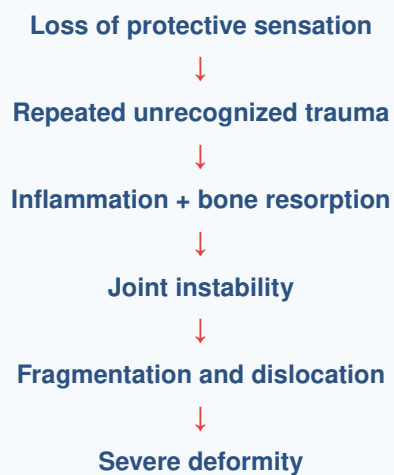
Other causes:

- Tabes dorsalis
- Syringomyelia
- Peripheral neuropathy
- Spinal cord disease
- Congenital sensory neuropathy

COMMON SITES

- Foot and ankle
- Midfoot
- Knee
- Hip
- Spine

PATHOPHYSIOLOGY



CLINICAL FEATURES

- Swelling
- Warmth
- Redness
- Instability
- Deformity
- Reduced pain despite severe structural damage
- Crepitus

- Altered gait

Foot:

- Characteristic rocker-bottom deformity may occur.

COMPLICATIONS

- Severe deformity
- Ulceration
- Infection
- Osteomyelitis
- Amputation
- Fractures
- Instability
- Loss of mobility

RADIOLOGICAL FEATURES

Early:

- Soft-tissue swelling
- Subchondral changes

Advanced:

- Joint destruction
- Fragmentation
- Subluxation
- Dislocation
- Bone sclerosis
- Osteophytes
- Debris
- Deformity

A classic description is “bag of bones” appearance in advanced disease.

CONSERVATIVE MANAGEMENT

Acute stage

- Complete/off-loading as indicated
- Total contact casting or appropriate immobilization
- Custom orthoses
- Regular skin inspection
- Glycemic control
- Foot care education

Chronic

- Custom footwear
- Orthoses
- Protected weight-bearing
- Gait training
- Fall prevention

SURGICAL MANAGEMENT

Indications:

- Severe deformity
- Instability
- Recurrent ulceration
- Failure of conservative treatment
- Severe structural destruction

Procedures:

- Arthrodesis
- Corrective osteotomy
- Exostectomy in selected cases
- Reconstruction
- Amputation for uncontrollable infection or non-reconstructable limb

CONNECTIVE-TISSUE DISORDERS

9. SYSTEMIC LUPUS ERYTHEMATOSUS — SLE

DEFINITION

Systemic lupus erythematosus (SLE) is a chronic systemic autoimmune disease characterized by autoantibody production and immune-mediated inflammation affecting multiple organs, including the skin, joints, kidneys, nervous system, blood and cardiovascular system.

CAUSES / RISK FACTORS

- Genetic predisposition
- Female sex
- Hormonal factors
- Environmental triggers
- Ultraviolet light
- Certain medications
- Infections may act as triggers

CLINICAL FEATURES

General

- Fatigue
- Fever
- Weight loss

Musculoskeletal

- Symmetrical joint pain
- Non-erosive inflammatory arthritis
- Morning stiffness
- Myalgia

Skin

- Malar rash
- Photosensitivity
- Discoid lesions

Other systems

- Oral ulcers
- Hair loss
- Renal disease
- Neurological manifestations
- Serositis
- Hematological abnormalities

DEFORMITIES

Usually non-erosive arthritis, but chronic disease may cause:

- Jaccoud arthropathy
- Ulnar deviation
- Swan-neck-like deformities
- Tendon abnormalities

These may be partially reducible, unlike fixed erosive RA deformities.

COMPLICATIONS

- Lupus nephritis
- Cardiovascular disease
- Stroke
- Seizures
- Anemia
- Thrombocytopenia
- Osteoporosis
- Avascular necrosis, particularly with prolonged corticosteroid exposure
- Infection

RADIOLOGICAL FEATURES

Usually:

- Soft-tissue swelling
- Osteopenia may occur

Unlike RA:

- Marked erosions are usually absent

X-rays may show:

- Jaccoud-type deformities without typical erosive destruction.

CONSERVATIVE / MEDICAL MANAGEMENT

- Patient education
- Sun protection
- Exercise
- Hydroxychloroquine in most appropriate patients

- NSAIDs for selected musculoskeletal symptoms
- Corticosteroids for active disease when indicated
- Immunosuppressive therapy
- Biologic therapy in selected patients
- Treatment of individual organ involvement

SURGICAL MANAGEMENT

Not usually directed at SLE itself.

May be required for:

- Severe joint destruction
- Avascular necrosis
- Fracture
- Tendon deformity

Procedures:

- Joint replacement
- Corrective surgery
- Fracture fixation

10. SCLERODERMA / SYSTEMIC SCLEROSIS

DEFINITION

Systemic sclerosis (scleroderma) is a chronic autoimmune connective-tissue disease characterized by vascular abnormalities, immune dysregulation and excessive collagen deposition, resulting in fibrosis and hardening of skin and potentially internal organs.

TYPES

1. Limited cutaneous systemic sclerosis

Skin involvement mainly distal to elbows/knees and face.

2. Diffuse cutaneous systemic sclerosis

More extensive skin involvement, including trunk and proximal limbs.

CAUSES / RISK FACTORS

Exact cause unknown.

Factors:

- Genetic predisposition
- Autoimmune mechanisms
- Environmental exposures
- Vascular dysfunction

CLINICAL FEATURES

Skin

- Skin thickening
- Tight shiny skin
- Loss of skin folds
- Reduced mobility

Hands

- Raynaud phenomenon
- Puffy fingers initially
- Sclerodactyly
- Digital ulcers
- Contractures

Face

- Tight facial skin
- Reduced mouth opening
- Mask-like appearance

Systemic

- Dysphagia
- Reflux
- Interstitial lung disease
- Pulmonary hypertension
- Renal crisis

DEFORMITIES

- Flexion contractures of fingers
- Sclerodactyly
- Reduced mouth opening
- Hand deformity
- Joint stiffness

COMPLICATIONS

- Digital ulcers
- Gangrene
- Pulmonary fibrosis
- Pulmonary hypertension
- Renal crisis
- Cardiac disease
- Esophageal dysfunction
- Severe functional disability

RADIOLOGICAL FEATURES

May show:

- Soft-tissue calcification
- Acro-osteolysis
- Joint-space changes
- Contractures

Chest imaging may demonstrate:

- Interstitial lung disease/fibrosis.

CONSERVATIVE MANAGEMENT

Medical

- Vasodilator therapy for Raynaud phenomenon
- Immunosuppressive therapy for selected systemic disease
- Proton-pump inhibitors for reflux
- Treatment of pulmonary hypertension
- Renal blood-pressure management

Physiotherapy

- ROM exercises
- Stretching
- Hand exercises
- Splinting when indicated
- Breathing exercises
- Aerobic conditioning
- Functional training
- Skin protection and education

SURGICAL MANAGEMENT

May include:

- Debridement of severe digital ulcers
- Management of gangrene
- Joint reconstruction in selected cases
- Surgical treatment of severe contractures

11. DERMATOMYOSITIS

DEFINITION

Dermatomyositis is an autoimmune inflammatory disease characterized by proximal skeletal muscle weakness and characteristic cutaneous manifestations.

CAUSES / RISK FACTORS

Exact cause unknown.

Possible associations:

- Autoimmune predisposition
- Genetic factors
- Environmental triggers
- In some adults, association with malignancy

CLINICAL FEATURES

Muscle

- Symmetrical proximal muscle weakness
- Difficulty climbing stairs
- Difficulty rising from a chair
- Difficulty lifting arms
- Difficulty combing hair

- Neck flexor weakness

Skin

- Heliotrope rash
- Gottron papules
- Photosensitive rash
- “Shawl” sign
- “V-sign”

Other

- Dysphagia
- Respiratory muscle involvement
- Fatigue

DEFORMITIES / FUNCTIONAL PROBLEMS

- Flexion contractures
- Poor posture
- Muscle wasting
- Reduced mobility
- Difficulty with transfers
- Gait abnormalities
- Children may develop calcinosis.

COMPLICATIONS

- Severe muscle weakness
- Dysphagia
- Aspiration
- Respiratory failure
- Interstitial lung disease
- Calcinosis
- Malignancy association in some adults
- Disability

RADIOLOGICAL FEATURES

There are no characteristic plain X-ray findings.

MRI muscle may show:

- Muscle edema
- Inflammatory changes
- Muscle atrophy in chronic disease

Other investigations:

- Elevated CK
- EMG abnormalities
- Muscle biopsy in selected cases

CONSERVATIVE / MEDICAL MANAGEMENT

Medical

- Corticosteroids

- Immunosuppressive agents
- IVIG in selected cases
- Other immunomodulatory/biologic therapies as appropriate
- Sun protection
- Treatment of dysphagia and respiratory complications

Physiotherapy

- Gentle ROM
- Gradual strengthening after disease control
- Aerobic conditioning
- Functional training
- Energy conservation
- Respiratory exercises where indicated
- Avoid aggressive exercise during uncontrolled active myositis.

SURGICAL MANAGEMENT

Usually not required for the primary disease.

May be required for:

- Severe calcinosis
- Contractures
- Secondary complications

12. POLIOMYELITIS

Important: Poliomyelitis is primarily an infectious viral disease of the nervous system, not a connective-tissue disease. It is included here because it is part of the requested syllabus.

DEFINITION

Poliomyelitis is an acute viral infection caused by poliovirus, primarily affecting the anterior horn cells of the spinal cord and sometimes the brainstem, resulting in flaccid muscle weakness or paralysis.

CAUSE

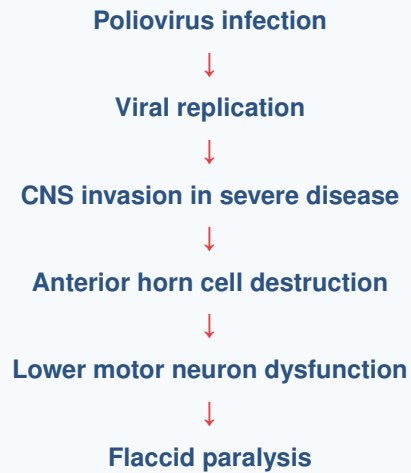
Causative organism:

- Poliovirus

Transmission:

- Mainly fecal-oral route
- Occasionally oral-oral transmission

PATHOPHYSIOLOGY



CLINICAL FEATURES

Prodromal

- Fever
- Headache
- Malaise
- Sore throat
- Vomiting

Paralytic disease

- Acute asymmetric weakness
- Flaccid paralysis
- Reduced muscle tone
- Reduced/absent reflexes
- Muscle wasting
- Fasciculations may occur
- Sensory function usually preserved

Respiratory involvement

- Intercostal weakness
- Diaphragmatic weakness
- Respiratory failure

DEFORMITIES

Long-term sequelae may include:

- Equinus deformity
- Knee flexion deformity
- Hip deformity
- Scoliosis
- Limb shortening
- Foot deformity
- Joint contractures

COMPLICATIONS

- Respiratory failure
- Aspiration
- Permanent paralysis
- Muscle wasting
- Contractures
- Skeletal deformity
- Post-polio syndrome years later
- Reduced functional independence

RADIOLOGICAL FEATURES

X-rays are not diagnostic of acute poliomyelitis.

Chronic cases may show:

- Muscle wasting-related osteopenia
- Limb shortening
- Deformities
- Scoliosis
- Joint contractures

CONSERVATIVE MANAGEMENT

Acute

- Supportive medical care
- Respiratory support
- Pain control
- Prevention of contractures
- Appropriate positioning

Rehabilitation

- ROM exercises
- Muscle strengthening of functioning muscles
- Orthoses
- Gait training
- Energy conservation
- Assistive devices
- Respiratory physiotherapy where indicated

Post-polio syndrome

- Pacing
- Energy conservation
- Avoid overuse weakness
- Appropriate low-to-moderate exercise
- Functional adaptation

SURGICAL MANAGEMENT

For residual deformities:

- Tendon transfer
- Corrective osteotomy

- Arthrodesis
- Limb-length procedures in selected cases
- Spinal surgery for severe scoliosis
- Orthopaedic reconstruction

13. MIXED CONNECTIVE-TISSUE DISEASE — MCTD

DEFINITION

Mixed connective-tissue disease (MCTD) is an autoimmune connective-tissue disorder characterized by overlapping clinical features of systemic lupus erythematosus, systemic sclerosis, polymyositis/dermatomyositis and rheumatoid arthritis, usually associated with high levels of anti-U1 RNP antibodies.

CAUSES

Exact cause is unknown.

Associated with:

- Genetic predisposition
- Autoimmune dysregulation
- Environmental triggers

CLINICAL FEATURES

Musculoskeletal

- Polyarthrititis
- Arthralgia
- Myalgia
- Proximal muscle weakness

Vascular

- Raynaud phenomenon

Skin

- Puffy fingers
- Sclerodactyly-like changes
- Rash

Other

- Dysphagia
- Esophageal dysfunction
- Interstitial lung disease
- Pulmonary hypertension
- Fatigue

DEFORMITIES

May develop:

- Joint deformity
- Contractures
- Sclerodactyly
- Muscle wasting
- Reduced hand function

COMPLICATIONS

- Pulmonary hypertension
- Interstitial lung disease
- Esophageal dysfunction
- Cardiac involvement
- Renal involvement, usually less prominent than in classic SLE
- Severe muscle weakness
- Functional disability

RADIOLOGICAL FEATURES

No single pathognomonic radiographic appearance.

May demonstrate:

- Joint-space narrowing
- Soft-tissue swelling
- Erosive arthritis in some patients
- Interstitial lung disease on chest X-ray/HRCT
- Muscle inflammation/edema on MRI

INVESTIGATIONS

Blood

- ANA
- Anti-U1 RNP
- CBC
- ESR/CRP
- CK
- Renal and liver function

Other

- Pulmonary function tests
- Echocardiography
- HRCT chest where indicated
- MRI muscle if myositis is suspected

CONSERVATIVE / MEDICAL MANAGEMENT

Treatment depends on the dominant manifestations.

May include:

- NSAIDs
- Corticosteroids
- Hydroxychloroquine
- Methotrexate
- Other immunosuppressive drugs
- Vasodilators for Raynaud phenomenon
- Treatment for pulmonary hypertension
- Treatment of interstitial lung disease

Physiotherapy

- ROM exercises

- Muscle strengthening
- Aerobic conditioning
- Breathing exercises
- Hand therapy
- Joint protection
- Energy conservation
- Functional training

SURGICAL MANAGEMENT

Usually not required for the primary disease.

May be considered for:

- Severe joint destruction
- Fixed deformity
- Contractures
- Digital complications
- Joint replacement in selected cases

HIGH-YIELD COMPARISON TABLE

Condition	Main pathology	Characteristic feature	X-ray / Imaging	Main management
Osteoarthritis	Cartilage degeneration	Activity-related pain	Joint-space narrowing, osteophytes, sclerosis	Exercise, weight management, analgesia, joint replacement
Rheumatoid arthritis	Autoimmune synovitis	Symmetrical inflammatory polyarthritis	Erosions + uniform joint-space loss	DMARDs + rehabilitation ± surgery
Ankylosing spondylitis	Axial inflammation	Inflammatory back pain	Sacroiliitis, syndesmophytes	Exercise + biologics ± surgery
Gout	Urate crystal deposition	Acute podagra	Punched-out erosions	NSAID/colchicine + urate lowering
Psoriatic arthritis	Immune-mediated inflammation	Dactylitis/psoriasis	Pencil-in-cup, new bone formation	DMARD/biologic + rehabilitation
Haemophilic arthritis	Recurrent hemarthrosis	Recurrent joint bleeding	Joint-space loss, cysts, enlargement	Factor replacement + physiotherapy
Still's disease	Systemic inflammatory disease	Fever + salmon rash + arthritis	Osteopenia/joint changes	Immunomodulation + rehabilitation
Charcot joint	Neuropathic destruction	Severe destruction with reduced pain	Fragmentation, dislocation, debris	Off-loading/orthosis ± reconstruction
SLE	Autoimmune multisystem disease	Non-erosive arthritis + systemic signs	Usually non-erosive	Immunomodulation
Scleroderma	Fibrosis/vascular disease	Skin tightening + Raynaud	Acro-osteolysis/calcification	Immunomodulation + functional therapy

Condition	Main pathology	Characteristic feature	X-ray / Imaging	Main management
Dermatomyositis	Inflammatory myopathy	Proximal weakness + rash	MRI muscle edema	Immunosuppression + graded exercise
Poliomyelitis	Viral anterior horn cell destruction	Flaccid asymmetric paralysis	No specific acute X-ray finding	Supportive care + rehabilitation
MCTD	Overlap autoimmune disease	Raynaud + mixed features	Variable	Immunomodulation + rehabilitation

VERY IMPORTANT EXAM DIFFERENCES

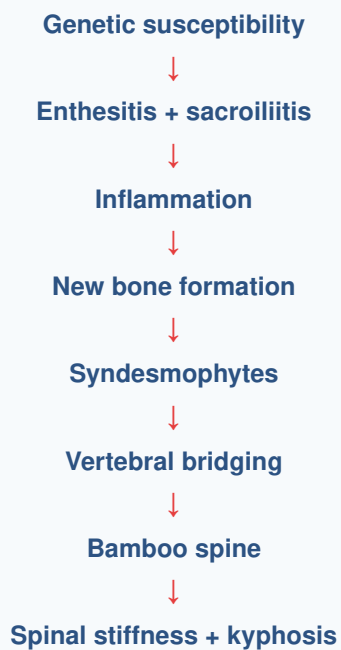
OSTEOARTHRITIS VS RHEUMATOID ARTHRITIS

Feature	Osteoarthritis	Rheumatoid Arthritis
Nature	Degenerative	Autoimmune inflammatory
Age	Usually older adults	Any age
Pain	Worse with activity	Inflammatory; often worse after rest
Morning stiffness	Usually <30 min	Usually >30–60 min
Joint involvement	Often asymmetric	Usually symmetrical
DIP	Commonly involved	Usually spared
MCP	Less typical	Common
X-ray	Osteophytes, sclerosis	Erosions, periarticular osteopenia
Deformity	Bony enlargement	Subluxation/deformity
Main drug treatment	Analgesia/anti-inflammatory	DMARDs

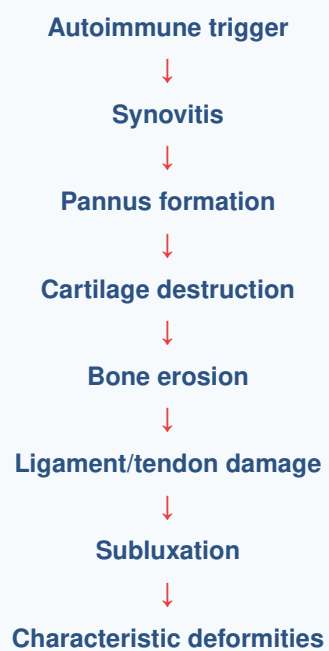
RA VS SLE ARTHRITIS

Rheumatoid arthritis	SLE
Usually erosive	Usually non-erosive
Persistent synovitis	Jaccoud arthropathy may occur
Marginal erosions	Systemic manifestations prominent
Joint destruction	—

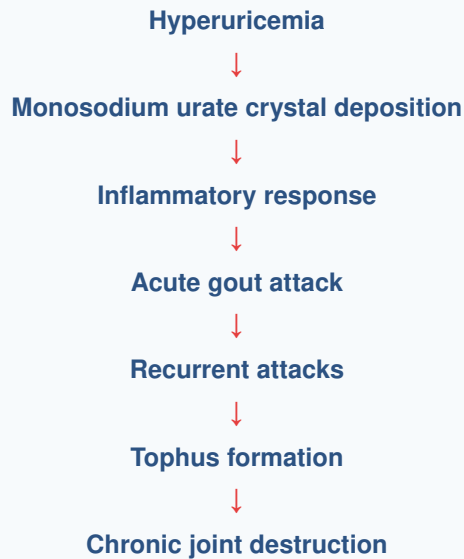
ANKYLOSING SPONDYLITIS — KEY FLOW CHART



RHEUMATOID ARTHRITIS — KEY FLOW CHART



GOUT — KEY FLOW CHART



STANDARD 20-MARK ANSWER STRUCTURE

For your university examination, use the following sequence for each condition:

1. **Definition — 2 marks:** Write a precise definition.
2. **Etiology/Causes/Risk factors — 2 marks**
3. **Classification — 1–2 marks:** Where applicable.
4. **Pathophysiology — 2 marks:** A flow chart gives good presentation.
5. **Clinical features — 3 marks:** Include Symptoms, Signs, Characteristic manifestations.
6. **Complications and deformities — 2 marks**
7. **Radiological features — 2 marks**
8. **Conservative/Medical management — 3 marks**
9. **Surgical management — 2 marks**
10. **Physiotherapy/rehabilitation + conclusion — 2 marks**

This format ensures that the answer covers the orthopaedic, medical and physiotherapy aspects expected in a BPT university examination.